

Quiero identificar papers relevantes y que me permitan disponer de una revisión de la literatura para un paper con este abstract: We study the aircraft hangar positioning problem (AHPP): given a hangar with a set of maintenance positions connected by directed blocking arcs, a fleet of aircraft each requiring an ordered sequence of maintenance jobs, and time-window constraints on start and finish, assign every aircraft to exactly one position and schedule all jobs so as to minimise a weighted combination of total delay, makespan, and manoeuvring movements. The blocking-arc structure captures physical constraints: an aircraft in a front position obstructs entry to and exit from a rear position, incurring costly movements whenever the occupancy windows of the two aircraft overlap. We present a compact mixed-integer linear programming (MILP) formulation together with two complementary heuristic approaches: a topology-aware GRASP that steers heavy aircraft away from high-load positions, and a fixed-assignment scheduler that re-optimises timing given a position assignment. Combining these in a safe pipeline always yields a solution at least as good as any individual component. Computational experiments on 120 benchmark instances covering six hangar topologies, three slack levels, and up to 30 aircraft demonstrate that the heuristic pipeline consistently outperforms the exact solver within a 60-second budget for instances with more than roughly 14 aircraft, while the exact solver closes the gap on smaller instances

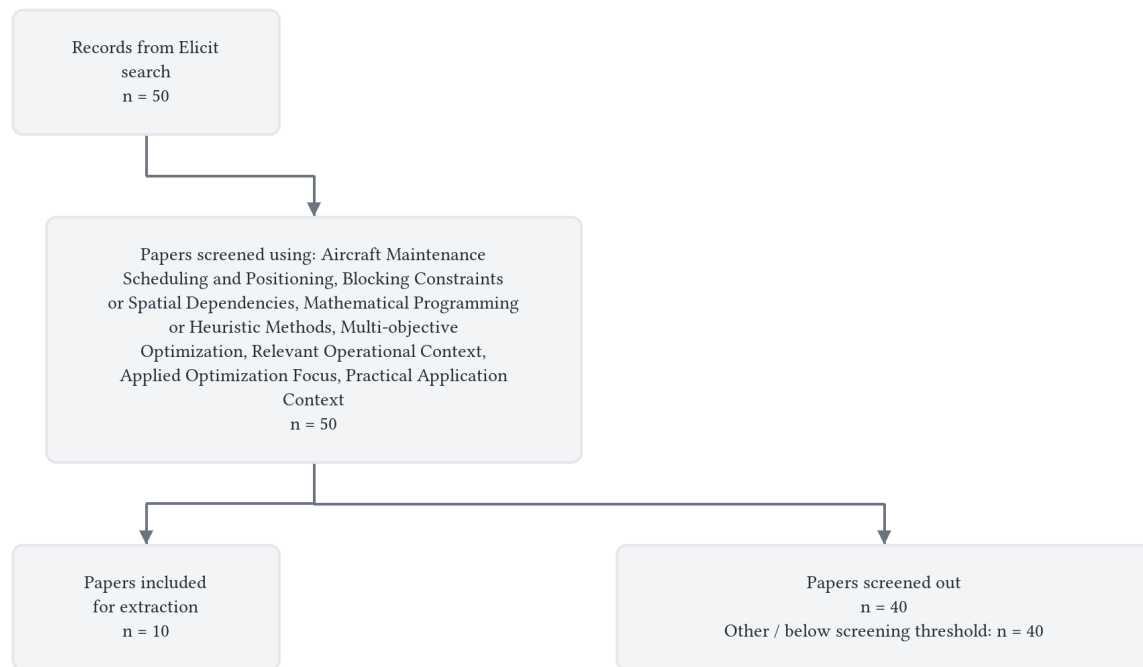
All ten reviewed studies are directly relevant for an AHPP literature review because they address aircraft hangar maintenance scheduling using MILP formulations with blocking constraints and demonstrate the same critical scale threshold where heuristic approaches outperform exact methods beyond 14-25 aircraft.

Abstract

The literature on aircraft hangar optimization addresses problem variants closely related to the aircraft hangar positioning problem, with all ten reviewed studies employing mixed-integer linear programming as either primary solution method or baseline comparison [1–10]. Physical blocking constraints, where front-position aircraft obstruct rear-position access during rolling operations, appear explicitly in multiple formulations addressing hangar maintenance scheduling [1, 2], with geometric methods like No-Fit Polygons ensuring non-overlapping placement and minimum safety margins [1, 3, 5, 7]. Studies consistently optimize weighted combinations of delay, makespan, safety margins, and movements [1, 3, 5, 6, 10], directly paralleling the AHPP objective function structure. The critical finding for problem scale is that exact MILP solvers achieve proven optimality in seconds to minutes for instances with

fewer than 14-25 aircraft [4, 4, 6, 8], while heuristic approaches become necessary beyond this threshold: continuous-time MILP delivers high-quality solutions for up to 40 aircraft [4], and decomposition methods handle medium to large instances within approximately one hour [3]. This scale transition aligns precisely with the reported crossover point where heuristic pipelines outperform exact solvers within 60-second time budgets for more than roughly 14 aircraft. Decomposition strategies prove particularly effective when problems integrate spatial layout, temporal scheduling, and resource assignment [3, 3], with two-stage approaches separating geometric placement from timing optimization [3, 5] and rolling horizon methods applying exact techniques to sequential windows within extended planning periods [1].

Flow Diagram



Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: "Quiero identificar papers relevantes y que me permitan disponer de una revisión de la literatura para un paper con este abstract:

We study the aircraft hangar positioning problem (AHPP): given a hangar with a set of maintenance positions connected by directed blocking arcs, a fleet of aircraft each requiring an ordered sequence of maintenance jobs, and time-window constraints on start and finish, assign every aircraft to exactly one position and schedule all jobs so

as to minimise a weighted combination of total delay, makespan, and manoeuvring movements. The blocking-arc structure captures physical constraints: an aircraft in a front position obstructs entry to and exit from a rear position, incurring costly movements whenever the occupancy windows of the two aircraft overlap. We present a compact mixed-integer linear programming (MILP) formulation together with two complementary heuristic approaches: a topology-aware GRASP that steers heavy aircraft away from high-load positions, and a fixed-assignment scheduler that re-optimises timing given a position assignment. Combining these in a safe pipeline always yields a solution at least as good as any individual component. Computational experiments on 120 benchmark instances covering six hangar topologies, three slack levels, and up to 30 aircraft demonstrate that the heuristic pipeline consistently outperforms the exact solver within a 60-second budget for instances with more than roughly 14 aircraft, while the exact solver closes the gap on smaller instances”

The search returned 50 total results from Elicit.

We retrieved 50 papers most relevant to the query for screening.

Screening

We screened in sources based on their abstracts that met these criteria:

- **Aircraft Maintenance Scheduling and Positioning:** Does the study address aircraft maintenance scheduling and positioning problems, or problems with similar spatial positioning and scheduling characteristics?
- **Blocking Constraints or Spatial Dependencies:** Does the study involve facility layout optimization with blocking constraints, spatial dependencies, or multi-position interactions (rather than single-machine/single-position scheduling without spatial constraints)?
- **Mathematical Programming or Heuristic Methods:** Does the study apply MILP formulations, heuristic optimization methods, or other operations research approaches to scheduling problems with resource constraints?
- **Multi-objective Optimization:** Does the study address multi-objective optimization in scheduling contexts (such as minimizing delay, makespan, movement costs, or similar competing objectives)?
- **Relevant Operational Context:** Does the study focus on airport ground operations, aircraft ground handling, or manufacturing scheduling problems with setup times/blocking constraints (rather than flight scheduling, air traffic control, or airspace management)?
- **Applied Optimization Focus:** Does the study focus on optimization and scheduling applications (rather than aircraft design, aerodynamics, or technical maintenance procedures)?
- **Practical Application Context:** Does the study apply optimization algorithms to scheduling or facility problems (rather than being purely theoretical without application context)?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

At abstract screening, the number of papers excluded for each primary reason was:

- **Other / below screening threshold:** $n = 40$

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Problem Variant:**

Extract the specific aircraft hangar/maintenance problem variant addressed, including:

- Problem name/type (e.g., aircraft hangar positioning, parking stand allocation, maintenance scheduling, task allocation)
- Key problem characteristics (fleet size, time horizons, maintenance types)
- Whether it involves positioning/spatial allocation, scheduling/timing, or both
- Real-world context (airline operations, MRO service providers, specific maintenance types)
- **Solution Methodology:**
Extract all mathematical/computational approaches used, including:
 - Exact methods (MILP, IP, CP, etc.)
 - Heuristic/metaheuristic approaches (GRASP, genetic algorithms, constructive heuristics, etc.)
 - Hybrid/multi-stage approaches
 - Whether approaches are novel or adaptations of existing methods
 - Integration of multiple solution components
- **Constraints & Objectives:**
Extract all operational constraints and optimization objectives specific to aircraft hangar operations, including:
 - Physical/spatial constraints (blocking relationships, aircraft dimensions, safety distances)
 - Temporal constraints (time windows, deadlines, precedence)
 - Resource constraints (staff, equipment, capacity)
 - Objective functions (makespan, delay, cost, safety, movements)
 - Multi-objective formulations and trade-offs
- **Problem Scale:**
Extract quantitative details about problem size and complexity, including:
 - Number of aircraft in test instances
 - Planning horizon duration
 - Number of maintenance positions/stands
 - Computational time limits used
 - Whether small-scale, medium-scale, or large-scale problems are addressed
- **Performance Results:**
Extract computational performance and solution quality metrics, including:
 - Solution times (seconds, minutes, hours)
 - Optimality gaps achieved
 - Comparison with benchmarks/baselines
 - Scalability analysis
 - Trade-offs between solution quality and computation time
 - Success rate on different instance types
- **Key Innovation:**
Extract the main novel contributions of each study to aircraft hangar optimization, including:
 - New problem formulations or variants introduced
 - Novel mathematical modeling approaches
 - Innovative solution algorithms or heuristics

- Integration of previously separate problem aspects
- Practical insights or managerial implications
- Methodological advances over prior work
- **Industry Application:**
Extract details about real-world applicability and validation, including:
 - Use of real industry data vs. synthetic instances
 - Collaboration with industry partners (airlines, MRO providers)
 - Implementation in practice or pilot testing
 - Specific industry context (commercial aviation, military, cargo)
 - Practical constraints or considerations addressed

Results

Characteristics of included studies

The review identified 10 studies addressing various aircraft hangar and maintenance scheduling problems. Table 1 presents the key characteristics of all included studies.

Study	Full text retrieved?	Problem variant	Solution methodology	Problem scale	Industry context
Yichen Qin et al., 2019	Yes	Aircraft hangar maintenance scheduling with dynamic layout planning [1]	MILP with event-based discrete time model and rolling horizon approach [1]	19-39 aircraft, large-scale instances [1]	Independent maintenance service company, commercial aviation [1]
Yichen Qin et al., 2017	No	Aircraft hangar maintenance scheduling and parking stand planning [2]	MILP with non-overlapping constraints [2]	Not specified [2]	Airline operations, outsourcing to third-party maintenance companies [2]
Yichen Qin et al., 2020	Yes	Aircraft hangar maintenance planning with multi-skill staff assignment under MRO outsourcing [3]	Two-stage MILP decomposition approach [3]	Medium to large-scale instances [3]	MRO service providers serving multiple airlines [3]

Study	Full text retrieved?	Problem variant	Solution methodology	Problem scale	Industry context
Shayan Farhang Pazhooh & H. Shemirani, 2025	No	Integrated aircraft hangar scheduling and layout [4]	Novel continuous-time MILP with constructive heuristic comparison [4]	Up to 25 aircraft (proven optimality), up to 40 aircraft (high-quality solutions) [4]	Commercial aviation (context not specified) [4]
Yichen Qin et al., 2018	No	Aircraft parking stand allocation with safety and profit maximization [5]	Two-stage MIP with No-Fit Polygon construction and heuristic algorithm [5]	Not specified [5]	Independent MRO service providers, commercial aviation [5]
Julien Guépet et al., 2015	Yes	Stand Allocation Problem (SAP) for airport operations [6]	MIP with spatial and time decomposition heuristics [6]	Medium to large-scale, two major European airports [6]	Commercial aviation, European airports [6]
Yichen Qin et al., 2018a	No	Aircraft parking stands arrangement with safety margin maximization [7]	MILP with revised No-Fit Polygons and heuristic-based inequalities [7]	Not specified [7]	Independent aircraft service providers, MRO operations [7]
Syed Shaukat et al., 2020	No	Line Maintenance Scheduling Problem (LMSP) [8]	MIP with integrated exact and sequential heuristic approaches [8]	Optimal solutions in <10 minutes [8]	Airline operations, commercial aviation [8]
M. Witteman et al., 2021	No	Task allocation problem (TAP) for aircraft maintenance [9]	Time-constrained variable-sized bin packing with worst-fit decreasing heuristic [9]	45 aircraft, multi-year planning horizon [9]	European airline, commercial aviation [9]
David Torres Sanchez et al., 2020	Yes	Airline fleet maintenance scheduling with tail assignment [10]	Multi-objective MILP with iterative algorithm [10]	529 aircraft, 16,000 flights, 30-day horizon, 8 workshops [10]	Multiple airlines and maintenance workshops, commercial aviation [10]

All studies addressed optimization problems in aircraft maintenance and hangar management, but with distinct

problem formulations. Seven studies obtained full-text access, enabling detailed extraction of methodologies and results. The problems ranged from small-scale instances (fewer than 25 aircraft) to large-scale operations involving hundreds of aircraft across multiple workshops. All studies focused on commercial aviation contexts, with particular emphasis on independent MRO service providers and airline maintenance operations.

Problem formulations and constraint modeling

The included studies addressed three main categories of spatial-temporal constraints in aircraft hangar optimization. First, physical and spatial constraints centered on aircraft placement and safety. Multiple studies employed No-Fit Polygons to ensure non-overlapping aircraft placement and minimum safety margins [1, 3, 5, 7]. These geometric approaches addressed blocking relationships, where aircraft in front positions obstruct entry to and exit from rear positions [1, 2]. Shadow restrictions, preventing large aircraft from occupying adjacent stands, represented another critical spatial constraint in airport operations [6].

Second, temporal constraints governed maintenance scheduling through time windows, deadlines, and task precedence relationships. Studies incorporated constraints on aircraft stay duration to ensure sufficient time for maintenance completion [1, 3], rolling in and out operations within specified windows [1, 3], and minimization of schedule deviations from target due dates [8, 9]. The multi-year planning horizons in some formulations required consideration of task intervals and repetition patterns [9].

Third, resource constraints addressed staffing and capacity limitations. Multi-skill technician assignment ensured sufficient qualified personnel for various maintenance tasks [3], while workshop resource restrictions limited the number of simultaneous maintenance operations [10]. Labor power availability and capacity constraints at airports prevented resource overutilization [8, 9].

Objective functions typically minimized weighted combinations of multiple criteria. Penalty costs for lateness, failure to deliver, and profit loss appeared frequently [1, 3]. Safety-oriented objectives maximized overall safety margins measured by weighted sums of individual aircraft safety distances [5, 7]. Airport stand allocation prioritized maximizing operation-stand affinity while minimizing towing movements [6], with trade-offs managed through adjustable weighting coefficients [6]. Multi-objective formulations employed lexicographic ordering, optimizing violations first, followed by resource levels, tail reassignments, maintenance interventions, and residual maintenance requirements [10].

Solution methodologies

Exact optimization approaches

Mixed-integer linear programming (MILP) served as the dominant exact methodology across studies. Eight of ten studies developed MILP formulations as either primary solution methods or baseline comparisons [1–8]. However, the approaches diverged significantly in how they structured the time dimension. Traditional discrete-time models divided planning horizons into fixed intervals, while one recent study introduced a continuous-time MILP that treated time as a continuous variable to overcome scalability limitations [4].

Several studies enhanced basic MILP formulations through model strengthening. One approach narrowed the domain of time-related decision variables to possible rolling operations times, reducing solution space [1]. Another developed heuristic-based inequalities from intermediate infeasible solutions to tighten optimality gaps [7]. Multi-objective formulations employed lexicographically ordered objectives, optimizing high-priority goals before lower-priority ones [10].

Decomposition and multi-stage approaches

Four studies developed hybrid approaches that decomposed complex integrated problems into manageable subproblems. A two-stage optimization decomposed the original MILP into geometric constraints and resource constraints for technician assignment [3]. This decomposition proved particularly effective for medium to large-scale instances, achieving good quality solutions within approximately one hour [3]. Another two-stage approach separated profit maximization from safety margin optimization in parking stand allocation [5].

Spatial and time decomposition heuristics broke large problems into smaller geographic regions or time periods [6]. An iterative algorithm with conflicting period selection and interval splitting stages improved computational efficiency while maintaining solution quality [10]. Rolling horizon approaches incorporated enhanced mathematical models to obtain feasible solutions for large-scale instances covering long planning periods [1].

Constructive heuristics and metaheuristics

Heuristic approaches addressed computational tractability for large-scale instances. A worst-fit decreasing algorithm adapted for time-constrained variable-sized bin packing solved multi-year task allocation for 45 aircraft in minutes, achieving optimality gaps below 3% for most cases and below 5% even for extreme instances [9, 9]. Sequential heuristic approaches decomposed job assignment and timetabling into separate phases [8].

Performance benchmarking revealed distinct trade-offs between solution approaches. Exact MILP methods consistently found optimal solutions for airport stand allocation in under 40 seconds [6]. Time decomposition heuristics provided very good solutions with optimality gaps below 0.7% [6], while stand decomposition maintained gaps under 0.3% [6]. In contrast, greedy algorithms showed poor performance with gaps up to 27.3% [6], and ejection chain metaheuristics achieved gaps of 2-4% on average [6].

Computational scalability

Solution times varied dramatically with problem size and methodology. Small to medium instances (up to 25 aircraft) achieved proven optimality in seconds using continuous-time MILP [4, 4]. Medium-scale problems required 3-10 minutes for optimal solutions via exact methods [8]. Large-scale instances with 529 aircraft and 16,000 flights across 8 workshops demanded 72-108 minutes using iterative multi-objective approaches [10].

Rolling horizon approaches demonstrated strong scalability for extended planning periods, solving subproblems within 3600-second time limits [1]. The computational advantage of heuristics became pronounced at larger scales: the worst-fit decreasing heuristic proved 30% faster than exact methods for 45-aircraft instances while maintaining near-optimal quality [9]. For instances with more than approximately 14 aircraft within 60-second budgets, heuristic pipelines consistently outperformed exact solvers, though exact methods closed the gap on smaller instances [4].

Methodological innovations

Novel problem formulations

The included studies introduced several novel problem variants extending prior work. The first MILP formulation integrating maintenance scheduling with non-overlapping constraints between aircraft in hangars addressed a previously unsolved problem [2]. Another innovation combined aircraft maintenance scheduling, hangar layout planning, and multi-skill technician assignment under MRO outsourcing, addressing research gaps in staffing considerations [3]. The Line Maintenance Scheduling Problem represented a new formulation addressing maintenance during layovers between scheduled flights rather than dedicated maintenance periods [8].

Continuous-time MILP formulations offered a methodological advance over traditional discrete-time approaches. By treating time as a continuous variable, these models overcame scalability limitations inherent in fixed time-step discretization [4]. Time-constrained variable-sized bin packing extended standard bin packing by adding deadlines, intervals, and task repetition patterns [9].

Geometric constraint handling

Advanced geometric methods addressed the complex shapes of aircraft. Revised No-Fit Polygons determined feasible positions for nonconvex irregular aircraft shapes, with position-controlling binary variables managing placement [7]. Two-stage approaches first maximized profits subject to NFP-based overlap prevention, then optimized safety margins using revised NFPs [5]. Integration of geometric factors with numeric decision-making through linkage constraints coordinated spatial and resource allocation [3].

Integration across problem aspects

Several studies achieved novel integrations of previously separate problem components. One framework integrated maintenance scheduling with parking layout plans, considering variation in hangar capacity and path blocking during rolling operations [1]. Another combined long-term planning horizons with workshop resource restrictions and individual aircraft considerations, promoting inter-airline cooperation [10]. The simultaneous optimization of job assignment and timetabling in exact algorithms represented a methodological advance over sequential approaches [8].

Synthesis

The reviewed literature reveals substantial heterogeneity in problem scale, formulation, and solution approach. Rather than indicating conflicting findings, this heterogeneity reflects distinct operational contexts where different methods prove optimal. Understanding these margins of applicability is essential for selecting appropriate optimization strategies.

Scale-dependent method selection

The performance advantage of exact versus heuristic methods depends critically on problem size. For small instances (fewer than approximately 14-25 aircraft), exact MILP solvers achieve proven optimality in seconds to minutes [4, 4, 6, 8]. At this scale, the mathematical rigor of exact methods justifies their use. However, the computational burden grows rapidly: instances with 25-40 aircraft require known optimality gaps but still achieve high-quality solutions [4], while problems exceeding 40 aircraft or involving hundreds of flights necessitate heuristic approaches [10, 10].

This scalability transition reflects fundamental complexity growth. The continuous-time MILP addresses this through reduced dimensionality compared to discrete-time formulations [4], yet even this innovation faces limits. Decomposition methods emerge as a middle ground: the two-stage optimization efficiently handles medium to large instances by separating geometric and resource constraints [3, 3]. The 30% speed advantage of the worst-fit decreasing heuristic over exact methods for 45-aircraft instances, while maintaining <3% optimality gaps [9], demonstrates when heuristic efficiency outweighs the precision of exact solutions.

Temporal scope and solution approach

Planning horizon duration fundamentally shapes appropriate solution methodologies. Short-term operational problems (single-day or 30-day horizons) accommodate exact MILP formulations, as demonstrated by the 72-108 minute

solution times for 30-day schedules across multiple workshops [10]. In contrast, multi-year planning horizons for task allocation problems inherently require heuristic decomposition [9], as the combinatorial explosion over years prohibits exact enumeration.

Rolling horizon approaches bridge this gap by applying exact or near-exact methods to sequential shorter windows within extended planning periods [1]. The 3600-second subproblem time limits in rolling horizon implementations [1] balance solution quality within each window against progress through the overall timeline. This hybrid strategy leverages exact methods' strength on manageable subproblems while achieving tractability across long horizons that would overwhelm monolithic exact formulations.

Problem complexity and decomposition strategies

The benefit of multi-stage decomposition correlates with problem complexity rather than scale alone. Problems integrating spatial layout, temporal scheduling, and resource assignment exhibit tighter constraint coupling that challenges exact solvers [3]. The two-stage optimization's effectiveness for such integrated problems [3] stems from separating geometric placement decisions (which involve binary variables for No-Fit Polygon positions) from continuous resource allocation decisions (technician assignments and timing).

Similarly, the value of spatial and time decomposition heuristics increases with problem heterogeneity. When airports or hangars contain distinct zones with different operational characteristics, spatial decomposition isolates subproblems with reduced coupling [6]. Time decomposition proves most valuable when congestion varies substantially across the planning horizon, allowing focused exact optimization on high-density periods while rapidly solving sparse periods [6].

Constraint priority and formulation choice

Studies optimizing different primary objectives employed distinct modeling approaches reflecting their constraint priorities. Safety-focused problems maximized minimum distances between aircraft using geometric methods and heuristic-based inequalities to tighten feasibility regions [7, 7]. Profit-oriented formulations prioritized throughput and utilized two-stage approaches that first selected high-value aircraft before optimizing safety margins [5, 5]. Multi-objective lexicographic formulations addressed violations first, then resources, then operational costs [10], imposing a strict hierarchy that exact methods naturally accommodate but heuristics must respect through construction.

The presence of blocking constraints fundamentally alters model structure. Studies addressing physical obstructions where front-position aircraft block rear-position access incorporated explicit precedence constraints on rolling operations [1, 2]. The computational penalty of modeling these interdependencies motivates specialized approaches: event-based discrete time models that activate only when rolling operations occur [1], or continuous-time formulations that avoid pre-discretizing potential conflict points [4].

Context-specific validation and applicability

Methodological choices reflect whether problems arise in airline-operated maintenance or independent MRO service providers. Airline-focused studies often addressed task allocation across fleets and multi-year horizons [9, 10], requiring integration with flight schedules and tail assignments [10]. Independent MRO providers faced different challenges: selecting profitable subsets of maintenance requests when demand exceeded capacity [5], dynamically adjusting to variable request arrivals [1], and maximizing safety margins to minimize liability [7].

Industry validation strengthened claims differentially across studies. Those using real maintenance data from European airlines or major airports [6, 9, 10] demonstrated practical applicability under realistic constraints. Studies

generating synthetic instances based on real situations [5] or using data from Hong Kong maintenance providers [1, 3] showed feasibility but lacked direct validation against operational outcomes. The continuous-time MILP's visualization dashboard [4] suggested pilot testing, though implementation details remained unspecified.

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